



Research paper

From surface to substance: Experiential learning to promote understanding of ChatGPT for K-12 lesson planning

Woonhee Sung^{a,*}, Heejung An^b^a School of Education, The University of Texas at Tyler, 3900 University Blvd, BEP 243, Tyler, TX, 75799, USA^b Department of Educational Leadership and Professional Studies, William Paterson University of New Jersey, 300 Pompton Road, Wayne, NJ, 07470, USA

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ABSTRACT

This study designed and implemented the “Re-Do” activity – a structured experiential learning intervention incorporating critical reflection that guides teachers through redeveloping their lessons using ChatGPT. The goal was to investigate how this intervention transformed 13 inservice teachers' understanding of AI integration at two U.S. state universities. Through inductive thematic analysis, findings revealed that teachers developed critical perspectives on integrating ChatGPT in education. Participants highlighted three key insights: the necessity of teacher expertise when evaluating AI-generated content, the importance of effective prompting strategies, and AI's potential to reduce lesson planning workload. The use of experiential learning facilitated teachers' progression through initial acquisition to specialized understanding, and finally toward integration of AI tools with a critical perspective. Results suggested that guided learning cycles and critical reflection promote informed technology integration that balances both technological capabilities and pedagogical expertise and is applicable across teacher education, preparation programs, professional development, and K-12 settings.

Major aspects of teachers' workloads, such as lesson planning, rubric development, and designing teaching activities, have historically been challenging and time-consuming (Wang, 2024), often leading to stress and burnout. Given that stress and workload are closely related to job satisfaction and retention (Barmby, 2006; Cayupe et al., 2023), this is an area in need of closer attention, especially given an ongoing lack of teachers in the nation's schools. When thoughtfully used, AI technologies can help address this issue by reducing teacher workloads through constructive and critical support (Hashem et al., 2024; Watermeyer et al., 2024). For example, recent studies have examined the effectiveness of generative AI as assistive tools to reduce teachers' workload in special education (Goldman et al., 2024) or secondary schools (Hashem et al., 2024).

The development of artificial intelligence (AI) technology has expedited the creation of AI-powered platforms and tools such as the Generative Pre-trained Transformer (GPT; e.g., ChatGPT or Google DeepMind Gemini), AI-powered writing assistants (e.g., Grammarly), image-generation tools (e.g., DALL·E), and content-creation tools for teaching (e.g., Magic School AI, Diffit, SlidesAI). Further, large language model (LLM) and natural language-processing technologies are applied in numerous AI-driven tools in various fields and disciplines, including

in education.

A number of studies have explored the benefits, potential, and challenges related to AI use in education (Adeshola & Adepoju, 2023; Delello et al., 2024; Fitria, 2021; Lo, 2023; Mao et al., 2024; Wu, 2023). Notably, ChatGPT has attracted attention due to its potential to provide tailored resources, ideas, and data for both teaching and learning (Baidoo-Anu & Ansah, 2023; Yu, 2023). Studies have also examined teachers' perceptions toward AI use in lesson planning (Karaman & Goksu, 2024; Koraishi, 2023; van den Berg & du Plessis, 2023). Teachers' daily routine includes developing lessons, modifying them based on different needs and purposes, designing assessments, and providing feedback. Teachers are experienced in these areas; however, they are not proficient in leveraging AI tools. As a result, unfamiliarity with AI tools and how AI functions may affect appropriate uses of AI, including ethical concerns (Delello et al., 2024; Mouta et al., 2024). Thus, while AI tools have attracted much attention among the general public, many teachers – not understanding or experiencing what AI might be capable of doing for their job functions – feel uncomfortable when hearing statements such as “AI might replace teachers” and, consequently, develop negative attitudes toward AI, or even fear of losing their jobs (Bryant et al., 2020; Chan & Tsi, 2024; Kasneci et al., 2023).

* Corresponding author. School of Education, The University of Texas at Tyler, 3900 University Blvd, Tyler, TX, 75799, USA.

E-mail address: wsung@uttyler.edu (W. Sung).

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All of these issues point to the need for professional development or teacher preparation that promotes critical perspectives and uses of AI to enable teachers to derive maximum benefit from these new technologies. Hence, in order to encourage taking a critical lens toward using generative AI (in this case, ChatGPT) for lesson planning and assessments, the current study introduced an intervention, “Re-Do,” to help teachers redo their lesson plans using ChatGPT following a set of instructional steps that required a comparison between their original lesson and the updated version resulting from use of ChatGPT. As part of the intervention, teachers also responded to a set of reflection questions designed to promote critical evaluation of AI.

1. Theoretical framework: experiential learning

The study used a structured learning process to expose teachers to the generative AI tool ChatGPT to increase their perspectives about and awareness of how to implement the tool for future uses. This approach promotes self-assessment and reflective practice regarding adoption of Open AI tools such as ChatGPT. That is, the intervention not only introduces the tool but also scaffolds teachers’ post reflections with critical understanding about the tool, encouraging deeper engagement and ethical consideration.

The intervention design was guided by experiential learning theory developed by David A. Kolb (Kolb, 1984, 2014; Kolb et al., 2014), which views learning as a holistic and cyclical process grounded in experience, where learners adapt to new learning through a continuous interplay of four stages: experience, reflection, conceptualization, and experimentation (see Fig. 1). As such, the theory highlights learning as an ongoing transformation of experience into knowledge rather than focusing on the outcome (Kolb & Kolb, 2009). The model is particularly relevant for technology integration in education as it embodies an iterative approach to engaging learners’ prior experience and beliefs about learning (Kolb, 1984).

At the first step, Experiencing (Concrete Experience), learning begins through the use of senses and perceptions to engage with the present moment. Learners immerse themselves in a new situation, gathering information through direct experience – in the current study, a ChatGPT tool. The next step, Reflecting (Reflective Observation), involves looking back on the experience, considering what happened, and linking thoughts and feelings to develop a deeper understanding, thereby processing the experience from multiple perspectives. In the third step, Thinking (Abstract Conceptualization), the learner engages in analytical thinking, drawing broader conclusions, and forming general principles,

theories, or concepts based on their reflections. Finally, in Acting (Active Experimentation), the learner applies the concepts or theories they have developed through active experimentation.

Building on this foundational cycle, this study also drew from Kolb and Kolb’s (2009) experiential learning theory of growth and development, which extends the basic experiential learning cycle into a developmental framework. This model defines three growth stages that learners progress through: (a) acquisition, where individuals develop basic abilities and cognitive structures; (b) specialization, involving development of particular learning styles shaped by educational and organizational socialization; and (c) integration, where nondominant modes of learning are expressed and incorporated into practice. For teachers engaging with ChatGPT, this developmental perspective helps explain how they might progress from initial exposure (acquisition), through specialized understanding within their teaching context (specialization), to eventually integrating AI tools with a balanced perspective that harmonizes pedagogical expertise with technological capabilities (integration).

Our study enriched Kolb’s reflection stage by adopting transformative learning theory (Mezirow, 1991, 2000), which emphasizes critical reflection. According to Mezirow (2000), critical examination and reflection on one’s beliefs leads to transformative learning, potentially shifting one’s perspectives or frameworks. Genuine reflection involves “active, persistent, and careful consideration of any belief or supposed form of knowledge in light of the grounds that support it and the further conclusions to which it tends” (Dewey, 1933, p. 9). Such depth of reflection was crucial for this study, as engaging teachers with the AI tool might raise questions about AI technologies and their implication in pedagogical practice.

2. Literature review

2.1. Teachers’ perceptions of applying generative AI in education

Numerous studies have explored attitudes toward the use of generative AI in educational settings. Perceived potential benefits include promotion of personalized learning, generation of assessments, data-driven insights, lesson planning, grading, and communicating in students’ home languages (An et al., 2024; Baidoo-Anu & Ansah, 2023; Crompton et al., 2022; Delello et al., 2024; Frick, 2024; Johnson et al., 2021). Limitations and drawbacks involve generation of incorrect information, perpetuation of biases, oversimplified decision-making, and privacy concerns (Baidoo-Anu & Ansah, 2023; Frick, 2024; Johnson

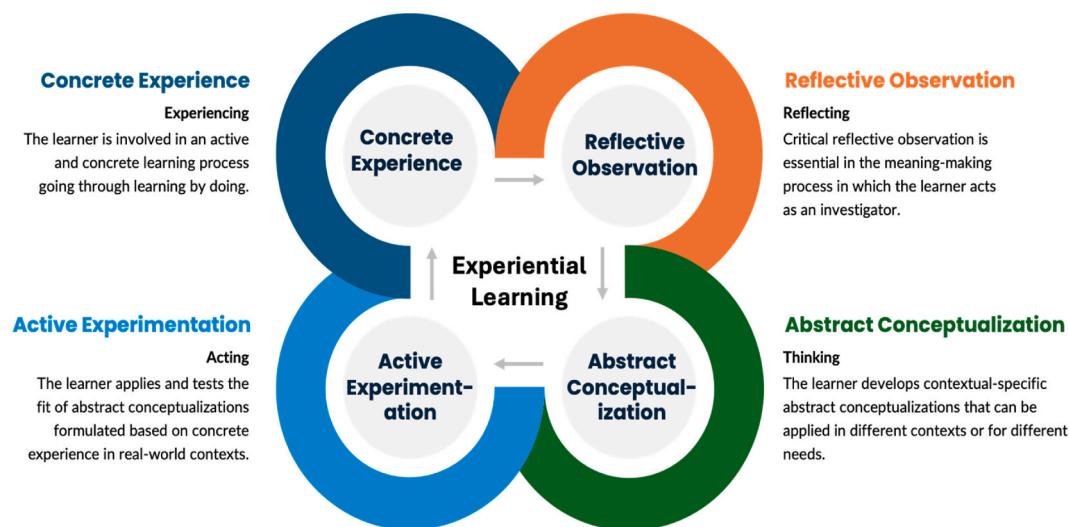


Fig. 1. Kolb's learning cycle. Adapted from Kolb (1984).

et al., 2021).

Among the daily tasks teachers engage in, lesson planning and updating of lessons requires considerable effort. AI tools can be beneficial in this regard; however, appropriate use is essential. To date, few studies have specifically examined how generative AI may assist with lesson planning or investigated educators' perceptions on the matter. Among existing study, for example, Lee and Zhai (2024) analyzed the lesson plans of 29 preservice elementary teachers in South Korea to assess how they integrated ChatGPT into science learning activities using a modified TPACK (Technological Pedagogical Content Knowledge) rubric. Findings showed that the teachers scored well on designing instructional strategies that included ChatGPT. However, they did not exploit all ChatGPT's capabilities, such as its potential for personalized learning and formative assessment.

Similarly, Uysal and Yüksel (2024) examined the integration of AI into language education, focusing on the perspectives of preservice English as a Foreign Language (EFL) teachers in Turkey regarding the effectiveness of AI-powered lesson plans. The participants created AI-generated lesson plans for teaching writing to fifth graders using a standardized prompt. Based on an evaluation rubric and qualitative analysis, findings highlighted the tool's effectiveness for engagement, appropriateness, and overall structure, while also noting challenges in differentiation and assessment.

Finally, according to a recent study by the RAND Corporation, 18 % of U.S. teachers reported using AI tools for teaching, with an additional 15 % having tried AI tools at least once (Diliberti et al., 2024). The survey revealed that educators who use AI explore it for a variety of purposes, including lesson planning, generating instructional materials, student assessments, and administrative tasks, as well as to provide personalized learning experiences. Among AI-using teachers, the most common applications were adapting instructional content to fit student levels (48 %), generating assessments (49 %), and creating lesson plans (41 %). Overall, the findings emphasized the need for more training and support to help teachers integrate AI effectively into their teaching practices. This conclusion was also emphasized in a systematic literature review by Chiu et al. (2023), which identified "insufficient knowledge of AI technologies among teachers" (p. 12) as one of the key challenges to AI integration in education. Thus, teachers may have a biased or incomplete understanding of AI tools, which may lead to inappropriate use.

In sum, while there is literature examining the benefits and limitations of using AI tools for lesson planning, there is a lack of empirical studies specifically investigating the development of teachers' critical lenses toward using AI tools for lesson planning.

2.2. Effects of experiencing learning with critical reflections

According to Dewey (1933), learning arises not from experience alone but from critical reflection, which allows individuals to derive meaning and guide future actions. This belief is central to transformative learning (Mezirow, 1990; Taylor, 2017), as reflection enables individuals to reassess their perspectives. Educators are, therefore, encouraged to engage in introspection through tools like reflective journals and facilitated discussions.

Empirical studies support the value of critical reflection in teacher development. For instance, Šarić and Šteh (2017) emphasized its importance but also identified barriers such as institutional constraints and limited reflective skills on the part of teachers. The researchers recommended fostering open-mindedness, institutional support, and targeted training programs to address these challenges. In the context of experiential learning, Girvan et al. (2016) found that when Irish second-level teachers engaged with new pedagogies as learners during curriculum reform, their beliefs and classroom practices changed meaningfully. Similarly, Wehbi (2011) showed that experiential teaching methods like simulations enhanced students' understanding of social justice and theory-practice links; however, the author cautioned about

potential discomfort without skilled facilitation.

To date, no studies have applied an approach to teacher training on the use of AI tools that combines critical reflection with experiential learning. This gap presents a unique opportunity to develop critical lenses for integrating AI into daily teaching practice, combining pedagogical innovation with reflective practice.

2.3. Design of the "Re-Do" intervention

Building upon Kolb's experiential learning theory (Kolb, 1984; Kolb et al., 2014) and Mezirow's (2000) transformative learning principles, the "Re-Do" intervention intentionally moves beyond surface-level technology exposure to create conditions for perspective transformation through structured experiences and guided reflection (see Fig. 2), thereby supporting not only skill development but also critical awareness and perspective transformation.

Unlike interventions that implement ChatGPT from the beginning of lesson development (Karaman & Goksu, 2024; Lee & Zhai, 2024; Powell & Courchesne, 2024; van den Berg & du Plessis, 2023), our approach positioned AI as a tool for enhancement rather than replacement – hence the name "Re-Do" (available via <https://tinyurl.com/2u8ch4dm>). This design choice aligns with transformative learning principles by creating cognitive dissonance – comparing one's established practice with an AI-enhanced alternative – which can trigger critical examination of assumptions about technology, pedagogy, and teacher expertise (i.e., Concrete and Reflective stages).

By guiding teachers through this structured process, the "Re-Do" intervention creates conditions for what Mezirow described as perspective transformation – a fundamental shift in how educators understand and engage with AI tools in their professional practice. That is, this design recognizes that meaningful integration of AI in education emerges not just from exposure to the technology but from critical engagement with its implications for teaching and learning. As such, the "Re-Do" framework supports teachers in moving beyond uncritical adoption or rejection of AI tools toward a more nuanced, critically informed stance that maintains teachers' agency while they explore technological affordances.

The study also draws on Kolb and Kolb's (2009) experiential learning theory of growth and development, which extends the basic experiential learning cycle into a developmental learning-focused framework. This model defines three learning growth stages that learners progress through: (a) acquisition, where individuals develop basic abilities and cognitive structures; (b) specialization, involving the development of a particular learning style shaped by educational and organizational socialization; and (c) integration, where nondominant modes of learning are expressed and incorporated into practice.

Development through these stages is characterized by increasing complexity in adapting to new contexts, which in the current study meant adapting to AI tools, and greater integration of learning modes as both educators and students in educational contexts. For teachers engaging with ChatGPT, this developmental perspective helps explain how they might progress from initial exposure (acquisition), through specialized understanding within their teaching context (specialization), to eventually integrating AI tools with a balanced perspective that harmonizes their pedagogical expertise with technological capabilities (integration).

The "Re-Do" activity was designed to facilitate this developmental journey by providing structured experiences that promote deep learning – learning that fully integrates the four modes of the experiential learning cycle. The model provides a framework for understanding how teachers develop from a surface-level engagement with ChatGPT (using it as an isolated tool) toward a more sophisticated integration where they can critically evaluate, adapt, and implement the technology as part of their ongoing professional practice.

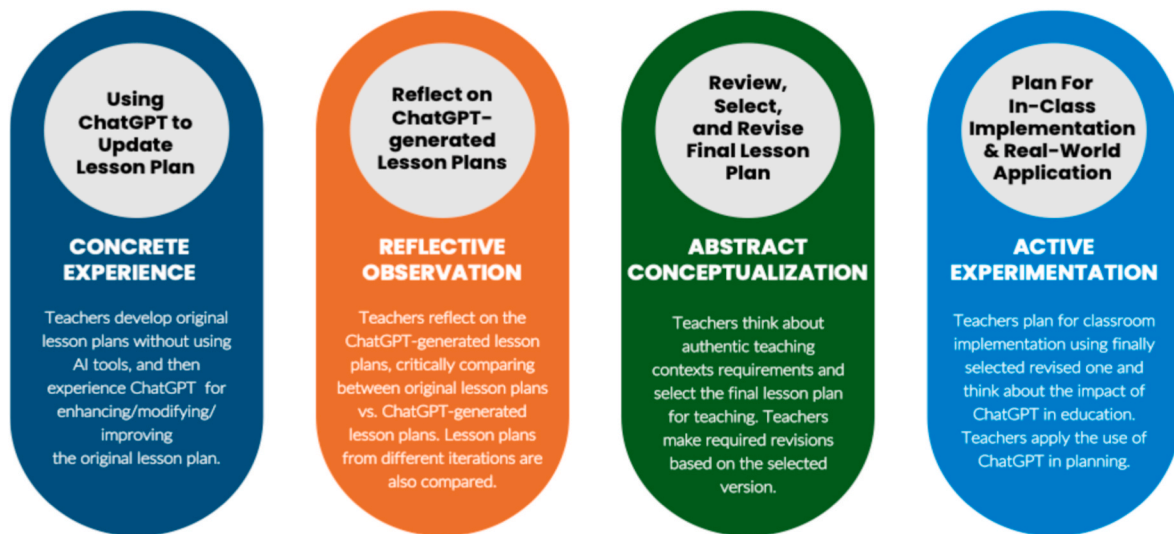


Fig. 2. The stages of intervention and steps adopted from the experiential learning model.

3. Purpose of the study and research questions

The study used a structured intervention called the “Re-Do” activity grounded in experiential learning and critical reflection to foster critical engagement with ChatGPT among a group of educators. Specifically, the intervention guided participating teachers in using ChatGPT to revisit and modify lesson plans they had developed independently. This sequential design enabled participants to directly experience the capabilities and limitations of AI in authentic educational tasks.

A primary aim of the study was to investigate how teachers’ perceptions of ChatGPT evolve through structured experiential learning and how educators develop critical reflection on implementing AI in their classroom practice. The following research questions guided the study.

- RQ1. How did educators’ perceptions of the benefits and risks of ChatGPT integration in K-12 education shift from before to after the “Re-Do” activity, considering both student and teacher perspectives?
- RQ2. What key themes emerged from educators’ open-ended responses regarding the practical advantages and challenges of using ChatGPT in educational settings?
- RQ3. What themes were found in the reflection questions about important takeaways, new discoveries, and understanding about lesson planning or assessment design reported by teachers after completing the “Re-Do” activity?

Rather than relying on abstract discussions about AI’s potential impact on education or providing lecture-based training about ethical AI use, the “Re-Do” activity provided concrete and interactive engagement with the tool in a context directly relevant to teachers’ professional practice. By engaging teachers in directly comparing their own work with AI-generated alternatives, the study created conditions where assumptions about both AI capabilities and teachers’ unique expertise could be examined and potentially reframed.

4. Methods

4.1. Participants

This case study was conducted at two state universities’ schools of education master’s programs located in the Southcentral ($n = 3$) and Northeastern regions ($n = 10$) of the United States. The authors of this paper were the instructors of the courses and requested voluntary participation from the students, who were inservice teachers. Thirteen

teachers participated, including 7 who explicitly identified as full-time teachers, 4 who provided specific grade levels/subject areas, 1 full-time student with past teaching experience, and 1 who did not identify their status. All participants were enrolled in master’s programs, with specializations primarily in MED in Curriculum and Learning: STEM/STEAM education ($n = 7$) and Curriculum & Instruction ($n = 5$); one unidentified.

Teachers who agreed to participate in the study completed pre- and post-survey and open-ended responses. Gender demographic included eight females and five males; the majority identified as White/Non-Hispanic ($n = 9$), followed by Hispanic/Latino ($n = 3$) and one not specified. Age distribution ranged from 24 to 56 years ($M = 38.08$, $Mdn = 36$), with the following distribution: 20–29 years ($n = 5$), 30–39 years ($n = 1$), 40–49 years ($n = 2$), and 50–59 years ($n = 4$). Teaching certification levels varied, including three teachers certified in elementary grades (K–6), three in both elementary and middle grades (K–8), one in elementary through secondary grades (K–12), one in middle grades (4–8), one in middle grades and special education (K–12), and three in secondary grades (6–12); one not recorded.

Teachers covered a broad range of grade levels, encompassing elementary through high school (grades 4–11), based on 10 recorded responses. Specifically, they taught at the elementary ($n = 2$; Grades 4–5), middle school ($n = 6$; Grades 6–8), and high school level ($n = 3$; Grades 9–11). Subject areas included mathematics ($n = 4$), science ($n = 3$), social studies ($n = 3$), reading/language arts ($n = 1$), and health ($n = 1$). Several participants ($n = 5$) taught multiple subjects or across multiple grade levels, such as “6–11 math, science, social studies,” “4- to 8-grade math,” “5th-grade RLA/SS,” and “Grade 5 science, health, and social studies.”

Participants had diverse teaching experiences, ranging from 0 to 22 years ($M = 9.41$, $Mdn = 6$). The teaching experience distribution included 0–5 years ($n = 5$), 6–10 years ($n = 2$), 11–15 years ($n = 1$), 16–20 years ($n = 2$), and more than 20 years ($n = 1$), representing a balanced mix of early-career, mid-career, and veteran educators. Finally, regarding prior experience with AI tools, eight participants reported previous AI use (61.5 %), while five had no prior AI experience (38.5 %). Specifically, for ChatGPT, the majority ($n = 8$, 61.5 %) had not used it before the intervention, while five had some prior experience with this tool (38.5 %).

4.2. Data collection and measurement

Data collection occurred in three phases. First, participants submitted their lesson plans and assessment assignments in their courses

without knowing they would be asked to modify them. Second, before implementing the “Re-Do” activity, participants completed a pre-intervention survey including consent information and demographic questions, such as teaching certificate status, academic major, and the grade levels currently taught or planning to teach. Participants were also asked about their prior experience with AI in general and ChatGPT specifically. This pre-intervention survey also included measures of participants’ perceived benefits and risks associated with ChatGPT use. Participants were prompted to consider benefits and risks from both student and teacher perspectives.

Third, after completing the “Re-Do” intervention, participants completed post-intervention measures identical to the pre-intervention measures of perceived benefits and risks, allowing for a nuanced analysis of how their views shifted as a result of the intervention. The post-intervention data collection also included three open-ended reflection questions administered simultaneously to elicit deeper insights into participants’ learning and perspective transformation after the “Re-Do” intervention. Specifically, participants were asked to (a) identify the most important takeaway regarding ChatGPT in K-12 education, (b) report new discoveries or realizations that emerged during the “Re-Do” project, and (c) describe any new learning regarding lesson and assessment design. By combining demographic data, structured pre- and post-intervention items, and rich qualitative reflections, the study aimed to capture the breadth and depth of educators’ evolving understandings of AI tool integration in educational practice.

4.3. Design of the study

The study employed a qualitative case study design, closely observing single groups before and after intervention responses and reflections. The “Re-Do” activity was embedded within existing graduate coursework without undergoing randomized controlled trials or comparison of alternative professional development formats. This design was selected for several reasons. First, experiential learning theory explicitly emphasizes the necessity of direct, hands-on engagement with tools through several iterations. In our study, the iteration started from developing individual lesson plans before exploring AI tools and incorporated contextually relevant, authentic practices, which cannot feasibly be replicated in alternative formats. Moreover, embedding the design within existing coursework for inservice teachers strengthened ecological validity, allowing teachers to experience the intervention within their authentic professional development context rather than in artificial conditions. Finally, as the research aimed to understand how structured experiential learning facilitates integration of AI in lesson-enhancing processes, using AI as enhancer rather than problem-solver, the qualitative analysis allowed us to capture the depth of perspective transformations in each stage and follow them to the end.

4.4. Data analysis

Participants’ demographic information and responses about previous AI/ChatGPT use were descriptively analyzed. For the open responses, the authors reviewed responses about the benefits and risks of ChatGPT use in education, as well as three post-reflective questions, following an inductive thematic analysis approach (Clarke & Braun, 2017).

The analysis process consisted of several systematic steps. First, each author read all responses multiple times to gain familiarity with the data. Then, they independently identified keywords, phrases, and concepts in each response, highlighting portions of text to determine codes that captured participants’ reflective thoughts and experiences. The coding process was iterative until a consensus was reached among researchers, with codes being refined as patterns emerged from the data. Related codes were grouped into broader themes based on their importance, similarities, and relationships that eventually evolved into overarching themes.

Throughout the process, the researchers met regularly to compare

their independently developed codes and themes, discussing any discrepancies until they reached 100 % agreement to achieve inter-rater reliability. The use of inductive thematic analysis was particularly valuable as it allowed meaningful patterns and insights to emerge naturally from participants’ reflections without being constrained by predetermined theoretical frameworks or expectations (Clarke & Braun, 2013, 2017).

After identifying the themes, the authors counted how often they appeared across the dataset in order to analyze frequencies to support theme salience (Guest et al., 2012) and to observe any differences in teachers’ perceived benefits and risks as reflected in their open responses and in post reflections. The themes and codes identified in the open responses were interpreted based on experiential learning theory and growth and development learning theory (Kolb, 1984; Kolb & Kolb, 2009) to understand participants’ developmental progression through the learning stages.

5. Findings

5.1. Shifts in perceived benefits and risks based on role

Before and after the intervention, participants were asked to consider the perceived risks and benefits of using ChatGPT from two distinct perspectives: as a K-12 teacher and as a graduate student. This dual-perspective approach was intentional, as participants were students enrolled in a master’s program in education while also serving as current or past K-12 teachers. That is, participants were encouraged to reflect on their experiences not only as learners but also as educators responsible for shaping classroom practice in the era of AI technologies.

The analysis examined codes from the original responses, themes that emerged from the codes, frequency of codes/themes that distinguished between pre-intervention and post-intervention responses, benefits vs. risks, and teacher vs. student perspectives. In the following tables, the “Teacher” columns represent participants’ responses when considering their role as K-12 educators, while the “Student” columns represent the same participants’ responses when considering their role as graduate students.

5.1.1. Analysis of perceived benefits and risks based on student and teacher roles

The analysis of perceived benefits and risks collected before and after the “Re-Do” activity revealed five major themes: (a) Efficiency and Time Management, (b) Instructional and Learning Support, (c) Quality and Accuracy Concerns, (d) Academic Integrity and Learning Concerns, and (e) Professional and Implementation Factors. The first two themes primarily captured perceived benefits, while the latter three encompassed various concerns and risks, with some implementation factors representing both opportunities and challenges.

Efficiency and Time Management. This theme emerged based on the codes “Ease of use,” “Time efficiency/quickness,” and “General task support.” Table 1 lists the frequency of codes in pre/post-intervention responses based on teacher and student perspectives. Before the intervention, both teacher and student role-based responses frequently mentioned time efficiency as benefit codes ($n = 10$) and showed a slight increase in the post-intervention ($n = 13$), reflecting that the perception about efficiency remained similar. The codes “Ease of use” and “General task support” were also found in both pre/post and teacher/student roles-based responses. The theme about efficiency was cited 37 times; however, no significant pattern was found between pre/post or between roles.

Instructional and Learning Support. This theme was identified based on the codes “Idea generation and drafting,” “Detailed and alternative explanation,” “Lesson improvement and refinement” and “Efficiency in Grading/feedback.” Table 2 lists the frequency of codes found in pre/post-intervention responses based on teacher and student roles. As illustrated, before the intervention, participants mainly viewed

Table 1

Codes for efficiency and time management theme and counts.

Codes	Before Intervention			After Intervention			Grand Total
	Teacher	Student	Pre N	Teacher	Student	Post N	
Ease of Use	3	2	5	3	1	4	9
Time Efficiency/Quickness	6	4	10	6	7	13	23
General Task Support	1	2	3	1	1	2	5
Total	10	8	18	10	9	19	37

Table 2

Codes for instructional and learning support theme and counts.

Codes	Before Intervention			After Intervention			Grand Total
	Teacher	Student	Pre N	Teacher	Student	Post N	
Idea Generation and Drafting	2	3	5	2	6	8	13
Detailed and Alternative Explanations	1	3	4	0	0	0	4
Lesson Improvement & Refinement	0	0	0	5	2	7	7
Efficiency in Grading & Feedback	1	0	1	1	2	3	4
Total	4	6	10	8	10	18	28

ChatGPT as a tool for generating ideas, offering alternative explanations, and supporting grading – functions commonly associated with general AI use. However, after the intervention there was a clear shift. Participants began to recognize specific aspects of ChatGPT's enhancement capabilities that they had not anticipated, such as its ability to suggest differentiation strategies and cross-curricular connections, rather than viewing it only as a general idea generator. Interestingly, the code for providing detailed and alternative answers disappeared in post-intervention, which may indicate that participants had developed a more refined understanding of how AI can support the planning of instruction. Despite these changes, however, generating ideas and lesson drafts remained the most frequently cited, highlighting ChatGPT's ongoing value in lesson design. The code "Idea generation and drafting" was the most frequently mentioned both before and after the intervention (13 mentions), with a notable increase in student responses post-intervention. Thus, both teacher and student perspective responses showed that ChatGPT was particularly useful for brainstorming and drafting lessons, reinforcing the tool's perceived strength in supporting teachers' instructional preparation tasks.

Quality and Accuracy Concerns. This theme was developed based on the codes related to concerns and risks, such as review of output needed, incorrect information, generic output concerns, and users' critical review needs. As seen in Table 3, the frequency of mentions increased dramatically from pre- ($n = 6$) to post-intervention ($n = 14$), indicating that participants not only learned ChatGPT's capabilities but also became more aware of limitations to be considered. It is particularly noteworthy that participants were initially only concerned about students using potentially incorrect output generated by ChatGPT; however, after the intervention, they recognized that teachers also need to be vigilant about output trustworthiness. Additionally, participants demonstrated increased concern about generic output quality ($N = 5$), which was not even discussed in the pre-intervention stage. The emergence of the "Critical user review needed" code in post-intervention responses, though minimal ($n = 1$), further suggests a growing

awareness of the need for critical evaluation. The increased number of codes related to the theme and codes appearing in post-intervention shows that teachers formed a more balanced understanding about ChatGPT, representing a healthy shift toward AI implementation in educational settings.

Academic Integrity and Learning Concerns. This theme encompassed codes related to plagiarism, reduced learning opportunities, masking authentic achievement, overreliance on technology, and diminished critical thinking skills (see Table 4). Analysis of participant responses revealed a striking pre-intervention asymmetry: Teachers expressed no concerns within this thematic domain ($n = 0$), while students' perspective showed substantial reservations ($n = 17$). Following the intervention, significant changes in code frequency were found, with teacher-perspective codes increasing from zero to 10. This shift suggests the experiential "Re-Do" activity catalyzed different perspectives toward the ethical uses of AI. The code about "Superficial learning demonstration" was absent in the beginning but appeared after the intervention in the teacher-perspective codes. Student-perspective codes maintained high frequencies throughout both phases, potentially indicating a persistent awareness of issues. The large increase in the number of teacher-perspective codes implies that the intervention appears to have facilitated a recognition of previously overlooked ethical dimensions of AI integration in education. These findings contribute to the growing discourse on ethics education around AI by demonstrating how experiential engagement influences educator perceptions, potentially leading to more nuanced implementation strategies that address both pedagogical opportunities and academic integrity challenges.

Professional and Implementation Factors. This theme emerged based on the keywords including "Privacy," "Complacency," "Not using teachers' expertise," "Focus more on students/reduce teacher load," "IEP/differentiation/ELL support," "Curriculum misalignment," "Lose teacher value concern," and "Less social interaction." The codes, count and patterns reported in Table 5 reveal a significant impact of the "Re-Do" activity – participants originally identified only two

Table 3

Codes for quality and accuracy concerns theme and counts.

Codes	Before Intervention			After Intervention			Grand Total
	Teacher	Student	Pre N	Teacher	Student	Post N	
Teacher Review & Oversight	2	0	2	0	1	1	3
Accuracy and Trustworthiness	0	4	4	2	6	8	12
Output Quality Issues	0	0	0	4	1	5	5
Critical User Review	0	0	0	0	1	1	1
Total	2	4	6	6	9	15	21

Table 4

Codes for academic integrity and learning concerns theme and counts.

Codes	Before Intervention			After Intervention			Grand Total
	Teacher	Student	Pre <i>N</i>	Teacher	Student	Post <i>N</i>	
Risk of Plagiarism and Lack of Original Thinking	0	8	8	3	7	10	18
Reduced Learning Opportunities	0	2	2	1	4	5	7
Superficial Learning Demonstration	0	1	1	3	0	3	4
Risk of Overreliance and Laziness	0	5	5	2	6	8	13
Decline in Critical Thinking, Creativity, and Self-Development	0	1	1	1	4	5	6
Total	0	17	17	10	21	31	48

Table 5

Codes and counts for teacher awareness of implementation outcomes.

Codes	Before Intervention			After Intervention			Grand Total
	Teacher	Student	Pre <i>N</i>	Teacher	Student	Post <i>N</i>	
Privacy and Data Security Concerns	0	0	0	1	1	2	2
Risk of Complacency	0	0	0	2	1	3	3
Not Utilizing Teacher Expertise in Lesson Development	0	0	0	1	0	1	1
Risk of Curriculum Misalignment	0	0	0	1	0	1	1
Reduce Teacher Loads (IEP, Differentiation, ELL Needs)	2	0	2	4	0	4	6
Concerns About Diminishing Teacher Value	0	0	0	2	0	2	2
Reduced Human and Social Interaction	0	0	0	1	0	1	1
Total	2	0	2	12	2	14	16

implementation-related considerations, while post-intervention data show a substantial expansion in recognized implementation factors, with total frequency increasing to 14. This dramatic shift was primarily driven by the teacher perspective. Other codes specifically addressed the collection of student- or teacher-perspective data from AI use leading to privacy concerns, potential loss of interactions, replacement of peer interactions, and concern about using the tool's ideas, not the teacher's. Two teachers expressed concerns about losing teachers' value in lesson planning and implementation, consistent with the "Risk of curriculum misalignment." However, teachers also demonstrated increased awareness of positive implementation factors, particularly the potential for ChatGPT to reduce administrative burdens and increase student engagement (3 post-intervention vs. 1 pre-intervention). The substantial increase in the number of implementation-related codes following the intervention indicates that direct experience with ChatGPT prompted participants, especially teachers, to consider more comprehensively how AI integration might affect not only instruction but also broader professional practice, educational institutions, and student-teacher relationships.

5.2. Themes from reflection questions on ChatGPT integration in K-12 education

To deepen understanding of educators' evolving perspectives on AI integration, participants responded to three reflection questions. The questions were asked after the intervention and were not sequential. Rather, each question was designed to elicit unique insights about different aspects of ChatGPT use in K-12 education, focusing on practical takeaways, new discoveries, and implications for lesson planning.

5.2.1. Important takeaway for educators using ChatGPT in K-12 education

The first reflection question prompted participants to consider practical applications and takeaways from their experience using ChatGPT for lesson planning. Analysis of responses revealed four predominant thematic categories: Need for Teacher Expertise, Efficient Lesson Planning, Concern as an Educator, and Recognition of Tool Limitations (see Table 6).

Need for Teacher Expertise. The most significant theme was the Need for Teacher Expertise when integrating ChatGPT in teaching. The code "Teacher expertise review/input needed" appeared with the highest frequency ($n = 7$), indicating a strong consensus about the

Table 6

Overall themes and codes found in "important takeaway" reflection question.

Theme	Codes	Q1 Count
Need for Teacher Expertise	Teacher expertise review/input needed	7
	Prompting strategy/multiple iterations needed	5
	Teacher expertise needed for implementation	2
	Total	14
Efficient Lesson Planning	Assist in creating lesson plan/component	6
	Provide ideas, alternative methods	4
	Useful for updating lessons	2
	Saving time	3
	Standard/cross-curricular integration	2
	Differentiation support	1
	Total	18
Concern as an Educator	Overreliance	4
	Impede learning and thinking skills	3
	Cautious use with clear purpose needed	2
	Student use concern related to academic integrity	1
	Less social interaction	1
	Total	11
Recognition of Tool Limitations	Generic output concern	4
	Error/trustworthiness output concern	2
	Not all answers, but a tool	1
	Total	7

importance of professional judgment when utilizing AI-generated content. Participants emphasized that while ChatGPT can generate lesson materials, educators must critically evaluate and refine these outputs based on their pedagogical knowledge. This is also related to the code concerning the quality of ChatGPT output being generic ($n = 4$). The code "Prompting strategy/multiple iterations needed" ($n = 5$) further underscores that participants recognized the importance of developing specific skills to guide AI tools toward desired outcomes. Teachers noted that initial outputs often required refinement through strategic follow-up prompts based on teachers' experience and knowledge, suggesting again the critical aspect of teachers' professional input.

For example, a sixth-grade science teacher first prompted ChatGPT by asking, "Create a 5E lesson (5E: Engage, Explore, Explain, Elaborate,

Evaluate) using this objective: Predict how geologic features of the Earth will be influenced in the future based on patterns of plate movement and Earth's physical properties," and then stated what had to be included in 5E based on the intervention instruction the authors provided. The follow-up prompts in the second and third trials asked for specific needs from the teacher's perspectives. The second prompt said, "Incorporate technology activities by using interactive simulations, virtual labs, multimedia presentations, digital modeling tools, and online research resources throughout the lesson. Encourage students to utilize technology for research, data analysis, and presentation of their findings. Ensure that the lesson is student-led by providing opportunities for collaborative learning, group discussions, and student presentations. As the facilitator, guide and support students as needed, but allow them to take ownership of their learning by actively participating in exploration, explanation, elaboration, and extension activities. In addition, could you add assessment ideas in the table per 5E stage?" The third prompt even included detailed analysis of learner populations with how to improve the table of activities provided in earlier iteration. In the fourth iteration, the teacher asked how to edit the table column titles and add the technology/non-technology component to the table.

A fifth-grade literacy teacher showed similar development from the initial prompt where the teacher asked for differentiating a lesson for emergent bilinguals and students reading below specified levels. Overall, participating teachers developed their prompts to a significant degree, by integrating their own contexts and advanced understanding/knowledge about content and students as they repeated the iterations.

Additionally, the code "Teacher expertise needed for implementation" ($n = 2$) reinforces the interpretation that educators viewed successful integration of AI-generated lessons as contingent upon their professional capabilities as teachers and facilitators who guide students' learning. That is, participants viewed AI as an assistive tool that enhances rather than replaces teacher expertise, suggesting that professional development focused on AI implementation should emphasize critical evaluation skills and effective prompting strategies alongside technical training.

Efficient Lesson Planning. This theme was frequently referenced, including the codes discussing the usability of ChatGPT in creating lesson plans, updating lesson plans, seeking ideas/alternative methods, integrating standard/curriculum, differentiating ideas, and time efficiency. As experienced through the "Re-Do" project, teachers valued ChatGPT's ability to generate lesson components such as objectives, activities, and assessments that could be customized or used to update existing lesson plans. The teachers also appreciated how the tool expanded their pedagogical repertoire by suggesting alternative or new approaches they might not have considered. This theme suggests that hands-on experiential learning with ChatGPT led to creation and improvement of their lesson planning process.

Concern as an Educator. Participants expressed concerns about AI implementation in educational contexts from educators' point of view. The codes "Overreliance" ($n = 4$) and "Impede learning and thinking skills" ($n = 3$) indicate educators' worries about becoming dependent on AI tools without developing their own learning and thinking skills. Further, the code "Cautious use with clear purpose needed" demonstrates that participants advocated for intentional integration rather than adoption for its own sake and found it essential for appropriate use. Other concerns referred to academic integrity and loss of social interaction (Crompton et al., 2022; Delello et al., 2024). These concerns, specific to educators, reflect a thoughtful acknowledgment of potential risks alongside benefits, considerations that are essential to developing critical understanding about the AI integration.

Recognition of Tool Limitations. The final theme captured participants' awareness of ChatGPT's technical and practical limitations. The most frequent code in this category, "Generic output concern" ($n = 4$), indicated that educators recognized that AI-generated content often lacks specificity and, therefore, needs teacher review within specific contexts. Participants also expressed "Error/trustworthiness output

concern" and "Not all answers, but a tool," indicating their realization that ChatGPT serves as one of the tools of their profession rather than *the* tool that could replace their profession. Participants' realistic understanding of AI's capabilities is critical in a tool adoption process that balances optimistic vs. pessimistic views. Recognition of these limitations suggests that educators approached AI integration with an appropriate level of critical awareness that would likely lead to more effective implementation.

5.2.2. New discoveries through completion of the "Re-Do" project

The second reflection question asked participants to identify any new discoveries they had made through the "Re-Do" project. While the variation of codes is similar to those of the first reflection question, the responses provided depth to the themes identified in the first question. Themes remain the same, as reported in Table 7, but the codes from the second reflection question are added. The codes "Standard/curricular integration" ($n = 2$), and "Assist in creating lesson plan/components" ($n = 3$) were additionally stated in the second round under the Efficient Lesson Planning theme. The two codes that highlight the need for teacher expertise and the importance of multiple/effective prompts were added under Need for Teacher Expertise theme. A new code emerged in this round, "Realize capability." Five teachers indicated they had not used ChatGPT or other AI tools in their lesson plans and had not considered it as a possible assistive tool. Although the code is simplified, teachers expressed how much they explored and found out the new possibilities of AI tools in supporting their instruction.

5.2.3. New realizations about designing lesson plans and assessments

The third reflection question prompted participants to reflect on broader considerations regarding implementation of ChatGPT in K-12

Table 7
Additional codes found in the second reflection question.

Theme	Codes	Q1 Count	Q2 Count
Need for Teacher Expertise	^b Teacher expertise review/input needed	7	+1
	^b Prompting strategy/multiple iterations needed	5	+1
	Teacher expertise needed for implementation	2	
Efficient Lesson Planning	Total	14	16(+2)
	^b Assist in creating lesson plan/component	6	+3
	Provide ideas, alternative methods	4	
	Useful for updating lessons	2	
	Saving time	3	
	^b Standard/cross-curricular integration	2	+2
	Differentiation support	1	
	^a Realize capability		+5
	Total	18	28(+10)
Concern as an Educator	Overreliance	4	
	Impede learning and thinking skills	3	
	Cautious use with clear purpose needed	2	
	Student use concern related to academic integrity	1	
	Less social interaction	1	
	Total	11	11(+0)
Recognition of Tool Limitations	^b Generic output concern	4	+1
	Error/trustworthiness output concern	2	
	Not all answers, but a tool	1	
	Total	7	8(+1)

^a new code(s) added in current (Q2) analysis.

^b Codes were added with more counts from the current analysis (Q2).

settings. Overall, the codes confirm the themes in the analyses of Reflection Questions 1 and 2. The major findings from the third reflection responses include realization about teachers' expertise for using and implementing AI, efficiency in lesson planning, and ethical concerns (see Table 8). A notable addition was made to the Need for Teacher Expertise theme by including six more codes on the necessity of teacher expertise and knowledge in reviewing input, prompting ChatGPT, and implementation. These codes reinforce teachers' consistent recognition and strengthened understanding of the need for professional judgment when implementing AI tools.

Furthermore, a significant number of codes related to Efficient Lesson Planning was added in this analysis. The most prominent code, "Assist in creating lesson plan/component" ($n = 6$), along with "Efficiency in planning/reduce teacher workload" ($n = 5$), highlights a significant shift in teachers' recognition of AI's practical value. Unlike in previous reflections where efficiency was noted but not emphasized as directly reducing workload, the third reflection data show teachers explicitly acknowledging how AI tools can alleviate instructional burdens. This is reinforced by moderate frequencies for "Generation of ideas and alternative strategies/methods," "Creating, updating, and adapting lesson plans," and "Time saving." The codes "Generic output concern"

and "Academic integrity concern" suggest that while these issues remained acknowledged, they were considered less urgent after the "Re-Do" project experience. This final reflection reveals a matured understanding where teachers prioritized AI's practical planning benefits while remaining aware of implementation considerations.

5.3. Findings related to demographic variations and open responses

Analysis of demographic variations revealed several interesting patterns; however, the small sample size limits definitive conclusions. Teachers with prior AI experience ($n = 5$) initially expressed more specific concerns about output quality and generic responses, while participants without prior experience ($n = 8$) mentioned general efficiency benefits. However, after the intervention, both groups came to realize teacher-specific perspectives and concerns. For example, STEM-focused teachers ($n = 7$) discussed prompting strategies and technical limitations, while Curriculum & Instruction teachers ($n = 5$) emphasized broader pedagogical implications and student learning concerns. Further, veteran teachers with 11+ years' experience ($n = 4$) expressed more concerns about traditional pedagogical practices being disrupted, while early-career teachers (0–5 years; $n = 5$) showed openness to AI integration with the concerns about professional development needs.

6. Discussion and implications

As illustrated, the "Re-Do" intervention facilitated participants' progression toward a better understanding of ChatGPT through the developmental stages outlined in Kolb and Kolb's (2009) framework: acquisition, specialization, and integration. The four phases of the experiential learning cycle served as the mechanism for facilitating this progression whereby participants advanced through the broader developmental stages, while making individual progression. Although our intervention did not include actual classroom implementation, findings showed perspective changes and increased understanding of ChatGPT use in education. Fig. 3 illustrates the extent of the impact of the "Re-Do" activity that implemented four experiential learning steps (light-blue small rectangle) into Kolb and Kolb's growth and development learning theory (blue large rectangle) (Kolb, 1984; Kolb & Kolb, 2009). Our thematic analysis provides evidence of collective movement through these stages at the group level. That is, teachers obtained basic understanding about the tool, specialized understanding as educators, and critical perspectives needed for implementation. Because the intervention did not include an actual implementation practice, the blue rectangle extends from acquisition to the middle of the integration stage.

As illustrated, at the Acquisition stage, participants gained initial exposure to ChatGPT through concrete experience, developing a basic understanding of its capabilities. Teachers formed their understanding of basic AI capabilities through hands-on experience with the technology, establishing the foundation for subsequent learning modes and development. This stage is characterized by what Kolb and Kolb (2009) described as a relatively undifferentiated approach to learning, where teachers were open to new experiences but had not yet developed specialized interactions with AI.

At the Specialization stage, participants began to develop educators' role-oriented ways of engaging with ChatGPT. According to Kolb's developmental theory, specialization involves the emergence of a particular and specialized understanding. Post-intervention responses showed this shift, with participants moving from generic AI benefits to educator-specific concerns about "teacher expertise needed for implementation" and "prompting strategy/multiple iterations needed." Some participants emphasized abstract conceptualization, focusing on theoretical implications and analytical concerns about AI use. Others gravitated toward active experimentation, prioritizing practical applications and technical implementations. Overall, at this stage, participants develop habitual learning patterns based on their educational backgrounds, career choices, and current job roles, resulting in a more

Table 8

Additional codes and count of new realizations about designing lesson plans and assessments.

Theme	Codes	Q1 Count	Q2 Count	Q3 Count
Need for Teacher Expertise	^b Need for teacher input/expertise	7	+1	+3
	^b Prompting strategy/Refinement through iterative prompting	5	+1	+1
	^b Expert guidance for implementation	2		+2
	Total: 22	14	16 (+2)	22 (+6)
Efficient Lesson Planning	^b Assistance with creating lesson plan/components	6	+3	+6
	^b Generation of ideas and alternative strategies/methods	4		+3
	^b Updating and adapting existing lessons	2		+3
	^b Saving time	3		+2
	^b Standard and cross-curricular alignment	2	+2	+1
	Differentiation support	1		
	^b Realization of instructional capability		+5	+2
	^a Efficiency in planning/reduced teacher workload			+5
	Total: 50	18	28 (+10)	50 (+22)
Concern as an Educator	Risk of Overreliance	4		
	Potential hindrance to critical thinking	3		
	Need for purposeful and cautious use	2		
	^b Academic integrity concerns (student use)	1		+1
	Reduced social interaction	1		
	Total: 12	11	11 (+0)	12 (+1)
Recognition of Tool Limitations	^b Generic or overly broad outputs	4	+1	+1
	Error and trustworthiness concerns	2		
	^b A tool for guidance, not definitive answers	1		+1
	Total: 10	7	8 (+1)	10 (+2)

^a new code(s) added in current (Q3) analysis.

^b Codes were added with more counts from the current analysis (Q3).

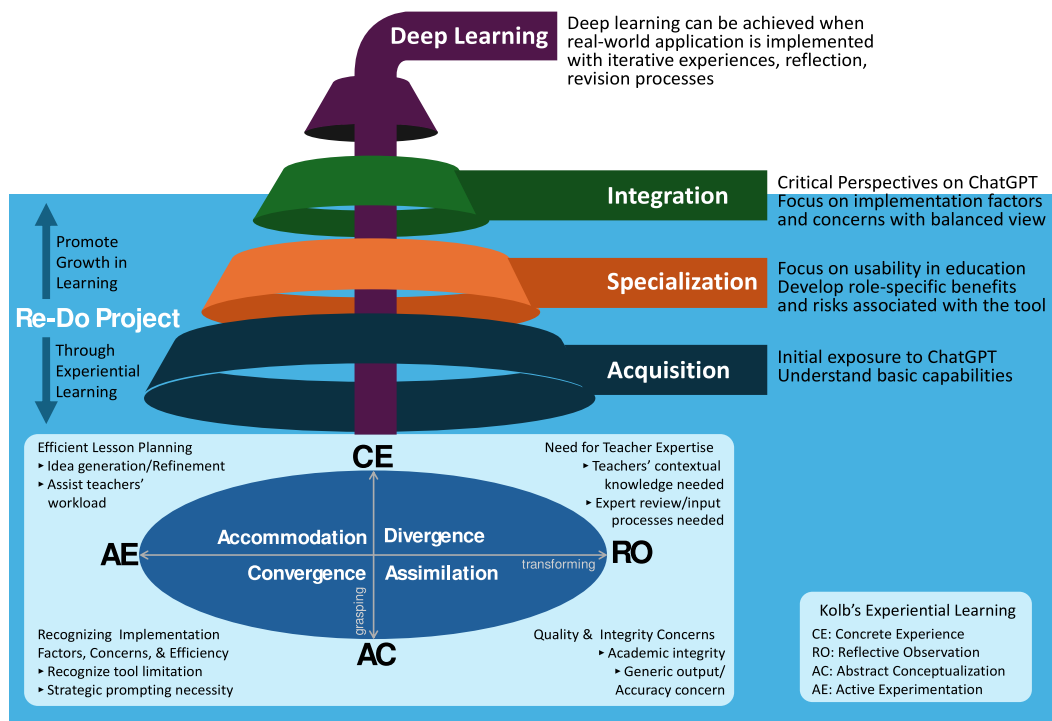


Fig. 3. Integration of themes in Kolb's experiential learning and growth in learning.

specialized but limited approach to new learning challenges (Kolb & Kolb, 2009).

Finally, at the Integration stage, participants developed integrative understanding – a level that could presumably have been strengthened if the intervention had required in-class implementation. Not all participants progressed through these stages uniformly, however – our analysis revealed collective patterns rather than individual developmental trajectories. Nevertheless, in our study, which provided concrete, reflect, conceptualization, and implementation thinking, teachers gained critical awareness of the tool's capability with a practical view toward implementation. Teachers specifically addressed instructional support capacity, as well as the importance of maintaining critical perspectives toward AI and teachers' expertise in utilizing the tool and keeping a balanced view between technical proficiency and ethical considerations. While our intervention did not include actual classroom use, the teachers' evolved perspectives, particularly their heightened awareness of AI limitations and emphasis on teacher expertise, likely represent critical prerequisites for successful classroom integration. Teachers who recognize the need for critical evaluation and strategic prompting are better positioned to model appropriate AI use for their students and avoid overreliance on AI-generated content. However, actual classroom implementation could introduce additional considerations such as student reactions, technical challenges, time constraints, and the need for real-time adaptation of AI outputs, areas of interest in future iterations of the intervention.

While our sample was limited to 13 teachers in U.S. graduate programs, the experiential learning framework and core themes – emphasizing teacher expertise, strategic prompting, and critical evaluation – align with fundamental pedagogical principles related to educational contexts in general. The “Re-Do” activity is fundamentally domain-neutral, as it centers on the universal practice of lesson planning, which exists across all subject areas and educational levels. That is, whether applied to mathematics, language arts, science, social studies, or other disciplines, teachers engage in the same core processes of setting objectives, designing activities, and creating assessments, making the intervention's structure of comparing original lessons with AI-enhanced versions broadly transferable across educational contexts

and subject matters. If applied in international settings, however, researchers and practitioners need to consider different levels of AI access, permissions, proficiency in digital literacy, and cultural attitudes toward technology integration.

The findings have practical implications for teacher education and professional development around AI literacy, suggesting that effective development should guide educators through a growth learning cycle promoting deep learning. Without task-specific guidance that emphasizes critical reflection, contextual relevance, and iterative practice of how to become critical users, teachers may feel pressured or perplexed about how to use AI tools. By structuring professional development that focuses more on reflective, iterative, and contextual experience rather than content-specific approach, institutions can facilitate sustainable technology integration, encouraging teachers to adopt AI as an intellectual partner, rather than passively accepting its outputs without evaluation or, at the other extreme, being overly dismissive of AI tools without understanding their potential affordances. In order to ensure broader applicability in diverse contexts, future research is encouraged on how cultural, technological infrastructure, and subject-specific factors influence teachers' AI integration experiences across different educational settings.

7. Conclusion

This study revealed that experiential learning with AI tools is crucial for developing teachers' nuanced understanding; theoretical discussions alone are insufficient. Educators need direct and concrete experience along with critical reflection to develop adequate understanding when adopting the new tool. The “Re-Do” project offered opportunities to think about implementation factors that are critical in adopting the new technology such as AI-driven tools. Recognition of these implementation considerations only emerged after the project, highlighting the significance of the intervention. It is also important to note that participants consistently emphasized that teacher expertise and strategies remain central to effective AI integration, viewing AI as complementary rather than replacement technology.

While the effectiveness of the intervention found significant shifts in

teacher perceptions, there are several limitations to this study. First, we acknowledge constraints due to the small sample size, which limits generalizability. Future studies with larger and more diverse samples can increase the transferability of the current intervention. However, although the participant group was small, their diverse backgrounds, certification areas, teaching experiences, and grade levels taught provided varied perspectives that enriched our findings and helped capture a range of insights that may inform future research on teacher engagement with AI tools.

Second, the study did not extend to actual classroom implementation, limiting our understanding of how teachers would apply their evolved perspectives in real-life teaching contexts. While participants developed critical awareness of and strategic approaches to using ChatGPT through the intervention, the study did not examine how these perspectives translate into actual teaching practice. Future studies can address these limitations by following teachers through classroom implementation phases with larger population. That is, longitudinal research could track how teachers apply their enhanced understanding of AI tools in authentic teaching contexts, documenting both implementation successes and challenges.

In closing, in providing detailed description of the “Re-Do” project, we hope that this methodology can be adapted for broader implementation. Future research would benefit from expanding the study to include more diverse educator populations or student perspectives across different institutional contexts.

CRedit authorship contribution statement

Woonhee Sung: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Investigation. **Heejung An:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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